

Stock Screener Application

1. Objective:

To develop an application to allow users to input some predetermined selection criteria, so the screener will scan through the whole database to filter a list of stock that met the selection criteria.

2. What are the pre-determined selection criteria?

- a. Market Capital Size = number of shares x latest price per share
- b. Revenue Growth Rate = latest revenue / revenue of prior year
- c. Profit Growth Rate = latest profit / profit of prior year
- d. Dividend Yield = latest year dividend in cents / latest share price in cents
- e. Dividend Growth Rate = latest full year dividend in cents / prior full year dividend in cents
- f. Price Earnings Ratio (P/E) = latest share price in cents / latest earnings per share in cents (= number of years required to recover your investment)
- g. Price to Book Ratio = share price in cents / book value per share in cents
- h. Debts to Equity Ratio = Total Debts / Total Equity
- i. OHLC – Daily Price (O-Open, H-High, L-Low, C-Close)
(default value is daily closing price)

3. In order to meet the above selection criteria, what are the data that the application need to collect?

- a. Exchange
- b. Stock Code
- c. Date
- d. Daily Price – Exchange, Stock Code, Date, OHLC (4 price data per day)
- e. Market Cap – Exchange, Stock Code, Latest Market Cap.
- f. Dividend declared per year in cents – Exchange, Stock Code, year, cents
- g. Revenue per year – Exchange, Stock Code, Year, Revenue
- h. Profit Before Tax per year – Exchange, Stock Code, Year, PBT
- i. Profit After Tax per year – Exchange, Stock Code, Year, PAT
- j. EBITA per year - Exchange, Stock Code, Year, EBITA

4. Need to create a table that show (to enable user to find out name of stock using the stock code.
 - a. Exchange
 - b. Stock Code
 - c. Stock Name

5. Filtering criteria can be a range from xxx to yyy and returning the results

Format of the outputs:

- a. List of selection criteria and value range,
- b. A table showing the result of the same criteria based on the selected stock with:

Exchange
Stock Code
Stock Name
Criteria 1, value
Criteria 2, value
Criteria 3, Value
Criteria 4, Value
Criteria 5, Value
....
....
....

6. Before making the final decision, the short-listed list can be uploaded to AI for recommendation with some other consideration like recent news, company forecast, other qualitative consideration. (optional steps)

I need the project developers (participating students) to go through the list and let the team knows what logistic is required and whether there will be problems in collecting data and any other potential issues. To provide feedback by 11 May 2026.